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Original Contribution

The impact of minimum wage policy on alcohol use disorder: a quasi-experimental study in South Korea

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Abstract

South Korea's 2018 minimum wage hike was examined for its impact on potential alcohol use disorders among affected individuals, using data from the Korea Welfare Panel Study (2015-2019). The study sample was restricted to workers aged 19-64 employed over the study years. The treatment group was identified as those below minimum wages, and the control group as those earning more than minimum wages in 2016-2017 ($n = 3117$ control, $n = 578$ treatment). Using outcomes derived from the Alcohol Use Disorders Identification Test, our results from difference-in-differences models showed that the 2018 wage hike was linked to a 1.9% increase in the "high risk" of alcohol use disorder and a 3.6% rise in hazardous consumption in the treatment group. Notably, the effects were more pronounced among men and those aged 50-64. Additionally, we confirmed that the spillover effects extended to workers earning up to 20% above the minimum wage. This study underscores the unintended substance use risk of minimum wage policies in the East Asian context. As wage policies are implemented, integrated public health campaigns targeting at-risk groups are required.

Key words: minimum wage; alcohol; substance use disorder; audit; difference-in-differences; quasi-experimental study; South Korea.

Introduction

Prior studies suggest that increases in minimum wages are associated with various health benefits, including reduced distress[1] and improved fruit and vegetable consumption[2]; however, their influence on obesity[3] and substance use behaviors, including smoking[4] and alcohol consumption, may be a concern. A comprehensive literature review explored the impact of minimum wage variations on health and health-related behaviors, including alcohol consumption.[5] Alcohol consumption usually rises with increased income[6]; however, increased income might also ease financial stress, potentially leading to a decrease in stress-induced alcohol consumption.[7] Despite numerous studies examining minimum wage and alcohol consumption, no studies to date have investigated whether the increased consumption might escalate to a point of potential alcohol use disorder, specifically the medical condition characterized by the inability to stop or control drinking despite adverse consequences.[8] While some previous studies examined heavy drinking,[9-11] they did not address alcohol use disorder. We aim to evaluate whether the large exogenous increase in the minimum wage through the 2018 national minimum wage policy in Korea has increased the risk of potential alcohol use disorder and harmful alcohol use among individuals affected by the policy, leveraging the methodological strengths identified in recent quasi-experimental studies[12] and employing a robust treatment-effect-on-treated (TEOT) approach.

Prior studies on minimum wage and alcohol consumption

An important element of causal inference studies on minimum wage and alcohol is based on devising an appropriate identification strategy to identify those who are likely affected by minimum wage and the counterfactual (ie, individuals not affected), and many prior studies have used age as an identification strategy. Previous studies were built with the assumption that teenagers[16-19] are more likely affected by minimum wage policy than adults.[13-16] A US study using the Youth Risk Behavior Survey found that a $1 increase in the minimum wage was associated with a 9% increase in binge drinking among teenagers, after controlling for potential confounders.[13] Another US study [14] found that a 10% minimum wage hike led to increased alcohol-related traffic deaths among teenage drivers by 5%-10%; however, the association was not observed among adult drivers. However, a study using the same data and study design,[15] but expanded from 1991-2013, concluded with null associations. However, the results of studies using teenagers and youths may not necessarily be generalizable to the entire low-income population since (1) the 2 groups have different alcohol use disorder risks[17], and (2) a larger proportion of teenagers' employment income is likely discretionary, which means that increases in minimum wages may lead to different alcohol consumption patterns. Another common method of identification relied on education, based on

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the assumption that those with less than college education are likely to be affected by the minimum wage. A study[10] using the US Behavioral Risk Factor Surveillance System (BRFSS) suggests that minimum wage increments did not significantly influence alcohol use among men but increased binge drinking among women aged 18-54 without a college degree. Similarly, increases in the minimum wage were inversely related to reported instances of alcohol consumption among Latino and Black men in the United States.[18] Another study[11] using BRFSS data found that increases in the state minimum wage were not associated with the number of alcoholic beverages consumed, probability of binge drinking, or being an at-risk drinker among those who have high school or less education. However, they found a significant increase in at-risk drinking among those without a high school degree. In addition, a study using data from BRFSS also found different effects (by gender) of minimum wage changes on binge drinking among low-educated young people (more for men and less for women).[9] In the East Asian context, a study[19] found that minimum wage hikes were linked to nonsignificant increases in the likelihood of drinking among workers with high school education or less; however, this study is limited by the exclusion of manufacturing sector workers. Prior studies using education as an identification method may be problematic since many people with high school education or less earn above minimum wage. For instance, in the United States, fewer than 5% of those with high school degrees or lower earn the federal minimum wage,[20] with only 28% of individuals between 25-50 years old with less than high school were considered "low-income."[21] In China, the majority of individuals with elementary and junior high education earn above minimum wage,[22] and in Korea, only 13%-15% earn minimum wage [23] despite 41% having education below a bachelor's in 2017.[24] Moreover, numerous earlier studies, particularly those using BRFSS data,[10,11,18] employed repeated cross-sectional designs, lacking the ability of prospective cohort designs to address unobserved individual heterogeneities,[7] a notable concern given the potential ties between persistent individual traits and excessive alcohol consumption [25].

On the other hand, a UK study that examined the impact of the National Minimum Wage policy in 1999, using panel data, found a nonsignificant reduction in the likelihood of drinking among low-educated workers [26]. However, the nonsignificant results may be due to the small sample size.

Minimum wage spillover effects

A limitation of education and age-based identification strategies is their inability to explore spillover effects. These effects are likely as firms aim to incentivize worker efforts following minimum wage hikes. To maintain relative wage differences, companies may need to raise wages for higher-skilled workers.[27] Additionally, employers may shift their hiring preferences towards skilled labor,[27] which leads to indirect wage increases among the high-skill group. Investigating spillover effects is vital for a comprehensive assessment of minimum wage increases. Evidence of spillover effects among those earning 5%-10% above minimum wage is common,[28] and effects on those earning up to 20% or higher above minimum wage are also observed in the United States and Korea.[29]

The Korean context

In 2018, Korea witnessed a significant increase in its minimum wage from ₩6470 to ₩7530, marking a rise of approximately 16.4%. This adjustment elevated Korea's ranking among the OECD (Organization for Economic Cooperation and Development) countries from 16th in 2016 to 14th in 2018.[30] Importantly, this increase shifted the minimum wage to the average wage ratio slightly above the OECD mean, presenting a valuable opportunity for a quasi-experimental study to examine the effects of exogenous income boosts on low-wage workers. The Minimum Wage Commission, tasked with the annual setting of the hourly minimum wage rate, plays a key role in this process. It consists of 27 members equally divided among representatives of employees, employers, and the public interest, with the latter often casting deciding votes amidst opposing views from the other 2 groups.[31] The determination of the minimum wage increase rate is influenced by partisanship, as the Minister of Employment and Labor, appointed by the government, holds the authority to decide the members of the Public Service Commissioners. Given the political dynamics, the substantial increase in 2018 stands out against a backdrop of smaller adjustments (ie, around 7% between 2016-17), making it a benchmark year for analyzing the impact of minimum wage policy change.

In 2018, South Korea's per capita alcohol consumption was 8.7 liters, aligning with the OECD average.[32] However, Korea has a deep-rooted culture of binge drinking and heavy episodic drinking. This culture is not just widespread but also celebrated in numerous facets of Korean life, from business meetings and office parties to social gatherings and traditional ceremonies, where alcohol plays a key role in fostering connections and community engagement.[32] These practices significantly drive the rise in alcohol use disorders, creating a major public health challenge. These practices have contributed to a rise in alcohol use disorders,[33] requiring more medical attention, and are associated with a range of negative consequences such as chronic diseases, injuries, suicides, criminal activities, and road traffic accidents.[34,35] While most prior studies examining minimum wage and alcohol consumption were limited by the use of age and education as the identification strategy, lack of prospective cohort data, and the use of single-item outcomes that did not evaluate the risk of alcohol use disorder,[36] we carried out a methodologically strong study by (1) employing a quasi-experimental study using wage as the identification strategy, (2) utilizing individual-level panel data, and (3) employing a multi-item questionnaire to assess the risk of alcohol use disorder, uniquely contributing to the minimum wage impact literature on substance use outcomes.

Method

Data

This study uses the 2015-19 Korean Welfare Panel Study (KOWEPS) data, waves 10-14. Notably, the prices of alcoholic beverages remained relatively stable during this timeframe.[37] The years 2020 and beyond were deliberately excluded from this study to avoid potential confounding effects of the COVID-19 pandemic. The KOWEPS is an annual panel survey that implements a multistage stratified probability sampling methodology. It is nationally representative, with data collected from 18 856 respondents spanning 7072 households, based on the Census data from 2005. Our study primarily aimed to examine the impact of minimum wage hikes on the risk of potential alcohol use disorders for working-age[19-64] employed workers. We excluded individuals who had no employment income in either 2016 or 2017. (See the Figure S1 for the included study cohort.) This decision was made to distinguish the impacts of wage augmentation from those of employment status change, supported by prior studies showing minimal influence of minimum wage increases on employment

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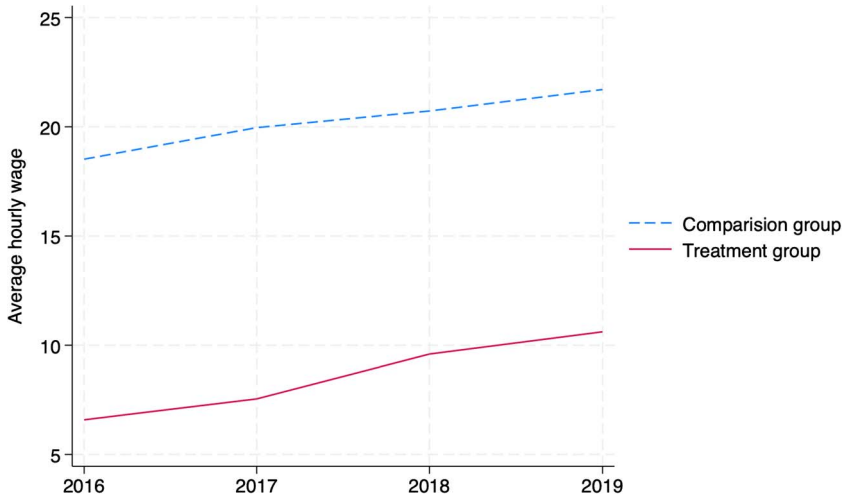

Figure 1. Average hourly wage comparison between target and comparison groups from 2016 to 2019.

in Korea.[38,39] Significantly, unemployment rates remained stable at around 3.7% before and after the implementation of the minimum wage increase.[40] Details about the KOWEPS are available on its official website.[41] This study is not subject to review by a research ethics board based on publicly available data.

Outcomes

Alcohol use disorders were assessed using the Alcohol Use Disorders Identification Test, Korean version (AUDIT-KR), and prior studies have provided evidence for the validity of AUDIT-KR[42-44]. The instrument has good reliability and consistency with the International Classification of Disease v10 definitions of alcohol dependence and harmful alcohol use.[44] The AUDIT comprises 10 questions (see Table S1). Based on the total scores, risk of alcohol use disorder is categorized into medium risk or above (≥8), high risk or above (≥16), and addiction likely (≥20). The AUDIT is also used to screen for supplementary outcomes concerning hazardous consumption, the presence of alcohol dependence, and the manifestation of alcohol-related harms. All 6 outcomes are used as dichotomous variables in our study.

Treatment assignment

The hourly earnings were calculated by combining the hourly income for temporary workers with the monthly income and usual total working hours for permanent workers. This method facilitated the identification of the treatment group (ie, at least 1 of the 2 years at or below ₩7530 in the 2 years preceding 2018) and the control group (individuals with hourly wages exceeding ₩7530 in 2016 and 2017). The Minimum Wage Act of Korea is applicable across all industries, employment statuses (ie, part-time, temporary, and foreign workers) and regions without exception.[31] Therefore, we applied the same identification strategy for all industries and regions. Figure 1 shows a notable increase in the average earnings among the treatment group in 2018 (from ₩7.5 k to ₩9.3 k), corresponding with a substantial minimum wage hike.

Given that prior research shows that minimum wage policies increased the wages of people earning slightly (5%-10%) more than minimum wage,[28] we formulated a second treatment group (ie, those with hourly wages below ₩ 8283, representing 110% of the 2018 minimum wage) in the 2 years leading up to 2018. This allowed us to more robustly explore the indirect impacts of the minimum wage changes.

Control variables

Any confounders that remain constant over the study period are accounted for by utilizing individual-level fixed effects.[45] Time-variant covariates (updated with each wave), including household size, health condition, marital status, home ownership status, and an indicator denoting disability, that may confound the association between minimum wage changes and alcohol are adjusted in the models. We did not include a control variable for alcohol prices since we saw only minor increases in alcohol prices in the study period (ie, there was only a 3% increase throughout. 2015-19).[46]

Statistical analysis

We used a difference-in-differences (DID) analytical approach, which examines the variations in outcomes before and after the implementation of treatment measures, contrasting the subjects who were subjected to treatment and those who were not. Our study diverges from prior cross-sectional research through the use of longitudinal microdata that facilitates the application of individual-fixed effects as a control mechanism for unobserved heterogeneity. Leveraging longitudinal data with individual-fixed effects, we probed the alterations occurring before and after within the same individual. DID analysis with binary outcomes is commonly modeled utilizing a linear model with fixed effect, in line with the existing literature,[7] because (1) contrasting parameter estimates across nonlinear models with varying covariate sets present complexities and (2) integrating fixed effects into logit and probit models has proven to be challenging.[47]

The main assumption of DID is that the control group reflects the counterfactual trend of the treatment group, which was validated by executing an individual-level 2-way fixed-effect regression that included an interaction term between the treatment assignment and the years in the pre-intervention period. Sampling weights supplied by the KOWEPS were used to generate representative estimates. Additional analyses included age and gender subgroup analyses. In addition, a prior study using 110%, 120%, and 130% of the minimum wage found the spillover effect of minimum wage changes on the health outcome,[26] and we followed their approach to establish our thresholds. The analysis was conducted using the STATA version 18. To check the robustness of our models, we have conducted multiple sensitivity analyses. The rationales, results, and interpretations are available in the supplementary file.

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Table 1. Baseline characteristics across treatment assignment, South Korea, 2015-2019 (unweighted $n = 3695$).

	Treatment group (col%)	Control group (col%)	χ2 test for difference in proportion between treatment and control groups
Number of participants	578 (100)	3117 (100)	
Gender			<0.001
Men	149 (25.8)	1830 (58.7)	
Women	429 (74.2)	1287 (41.3)	
Age			<0.001
19-29	99 (17.1)	229 (7.3)	
30-44	186 (32.2)	1708 (54.8)	
45-64	293 (50.7)	1180 (37.9)	
Homeownership			<0.001
Yes	289 (50.0)	1792 (57.5)	
No	289 (50.0)	1325 (42.5)	
Marital status			<0.001
Married	299 (51.7)	2026 (65.0)	
Unmarried	279 (48.3)	1091 (35.0)	
Disability			0.001
Yes	32 (5.5)	87 (2.8)	
No	546 (94.5)	3030 (97.2)	
Household size			0.082
1	39 (6.7)	217 (7.0)	
2	103 (17.8)	497 (15.9)	
3	180 (31.1)	857 (27.5)	
4 or more	256 (44.3)	1546 (49.6)	
Self-reported health			<0.001
Excellent, very good and good	450 (77.9)	2655 (85.2)	
Fair and poor	128 (22.1)	462 (14.8)	

Results

Table 1 displays the baseline characteristics ($n = 3117$). The treatment group (ie, those likely to be influenced by the minimum wage policy) has a higher percentage of women, individuals aged 45-64, renters, those who are unmarried, individuals with disabilities, solitary residents, and those in poor/fair health when contrasted with the control group over study years. Table 2 presents age-standardized incidence rates for alcohol misuse by gender per 1000 person-years over the study period. Overall, the control group had higher incidence rates than the treatment group. For example, in outcomes related to hazardous drinking, the treatment group recorded 114.75 incidents, whereas the control group had 148.41 incidents per 1000 person-years. Notably among men, the control group displayed higher incidence rates for all types of alcohol misuse. Meanwhile, women in the treatment group had higher incidence rates in the "addiction likely" risk categories than their women counterparts in the control group.

Table 3 shows the main results from the 6 fully adjusted DID regression models, focusing on the interaction between the pre/postindicator and treatment group status, where a significant and positive coefficient (of the interaction) captures the treatment effect (ie, additional risk on the treatment group due to the policy change over and above the control group). The TWFE models without spillover effect (left panel) show that the minimum wage hikes in 2018 increased the risk of "addiction likely" by 1.0% (95% CI, 0.0%-1.9%), risk of "hazardous" drinking by 4.2% (95% CI, 0.4%-7.99%). The right panel of Table 3 shows the 10% spillover effect analyses, where workers who earn up to 110% are in the treatment group (see also Figure S2). The minimum wage hikes in 2018 increased the probability of high risk by 1.9% (95% CI, 0.2%-3.6%), the risk of "addiction likely" by 1.2% (95% CI, 0.3%-2.0%), the risk of "hazardous" drinking by 3.6% (95% CI, 0.2%-7.1%), the risk of "dependence" drinking by 3.2% (95% CI, 0.0%-6.4%), and the risk of alcohol-related harm by 4.3% (95% CI, 1.4%-7.2%). Results for the 20% and 30% above minimum wage for spillover effects were overall not significant (see Table S4).

To establish that the parallel trend assumption is not violated, we show (1) the impacts of treatment status over time (in years) in Table S2 and (2) conducted F-tests showing that the coefficients of leading terms (before 2018) are not significant. Additionally, we produced the continuous AUDIT scale and proportion of hazardous drinking for treatment and control groups over time (Figure S3).

Results of subgroup analyses by gender and age groups are in Table S3. The effects of the minimum wage hike on alcohol disorders are more clearly observed in men and older age groups.[50-64] As with the main models, the policy effect was not significant for medium-risk outcomes but was more pronounced in severe outcomes, including high-risk and addiction likely. For additional spillover effects analyses, Table S4 shows elevated risks for high risk of alcohol disorder, "dependence," and "harm" for workers who earn up to 120% of the minimum wage; fewer significant associations were observed among those earning up to 130% of the minimum wage. In addition, findings from our sensitivity analyses (Table S5-Table S13) are overall consistent with our main findings.

Discussion

The 2018 minimum wage hike in Korea was associated with increased risks of alcohol use disorder (ie, shown in models for high risk and addiction likely), as well as harmful alcohol use (ie, shown in models for hazardous drinking and alcohol-related harms) among the treatment group. For example, our results suggest that a 13.3% increase in the real minimum wage led to a roughly 1.9% increase in the probability of high-risk alcohol

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Table 2. Age-standardized incidence rates for alcohol misuse by gender per 1000 person-years, 2015-2019.

	All (men and women)	Men	Women
Medium risk for alcohol use disorder[a]	<0.001	<0.001	<0.001
Treatment	42.90 (42.90-42.91)	78.71 (78.68-78.73)	29.24 (29.23-29.25)
Control	68.49 (68.48-68.50)	109.48 (109.46-109.50)	28.75 (28.75-28.77)
High risk for alcohol use disorder[b]	<0.001	<0.001	<0.001
Treatment	6.24 (6.23-6.24)	15.76 (15.75-15.77)	2.45 (2.45-2.46)
Control	10.84 (10.83-10.84)	18.47 (18.46-18.47)	2.97 (2.97-2.98)
Addiction likely[c]	<0.001	<0.001	<0.001
Treatment	2.72 (2.71-2.72)	5.96 (5.95-5.96)	1.44 (1.44-1.44)
Control	3.70 (3.69-3.70)	6.97 (6.96-6.97)	0.98 (0.97-0.98)
Hazardous consumption[d]	<0.001	<0.001	<0.001
Treatment	114.75 (114.74-114.76)	156.97 (156.94-157.00)	98.76 (98.74-98.77)
Control	148.41 (148.40-148.42)	185.57 (185.53-185.59)	107.77 (107.76-107.79)
Alcohol dependence[e]	<0.001	<0.001	<0.001
Treatment	27.23 (27.22-27.24)	53.20 (53.18-53.22)	18.76 (18.75-18.77)
Control	40.51 (40.50-40.52)	59.02 (59.00-59.03)	21.35 (21.35-21.36)
Alcohol-related Harm[f]	<0.001	<0.001	<0.001
Treatment	25.39 (25.38-25.39)	48.13 (48.11-48.15)	17.73 (17.73-17.74)
Control	40.64 (40.64-40.65)	59.90 (59.88-59.92)	21.33 (21.33-21.34)

[a]Medium risk: Alcohol Use Disorders Identification Test (AUDIT) score ≥8.
[b]High risk: AUDIT score ≥16.
[c]Addiction likely: AUDIT score ≥20.
[d]Hazardous: hazardous levels of consumption.
[e]Dependence: the presence or emergence of alcohol dependence.
[f]Harm: the manifestation of alcohol-related harm.

use disorder and 3.6% increase in the probability of hazardous consumption. In addition, subsample analysis shows the effects are more likely driven by men and older people. We also found spillover effects up to 110% of minimum wage.

Prior studies have focused on the association between minimum wage and alcohol consumption (either frequency or amount), binge drinking (typically based on a number of drinks in a single occasion), or drinking-related traffic mortality[14,15] rather than the risk of alcohol use disorder (ie, a medical condition marked by the inability to stop or control drinking despite negative consequences). Thus, one of the main contributions of this article is to clarify the impact of a minimum wage hike on the probability of alcohol use disorder, presenting a clinically significant outcome for intervention strategies. Another contribution is that our study design, employing a TEOT approach, allows us to estimate more precise policy effects. While the outcome in this study is different from the prior studies, comparing them may still help clarify the broader implications of minimum wage policies on alcohol-related outcomes. Three studies using the BRFSS all used repeated cross-sectional study design (limited by compositional change over time) to examine minimum wage impact on alcohol consumption and/or binge drinking. This first one did not find any significant policy impact on alcohol consumption or binge drinking for the general lower-income workers. However, it did observe significant increases in at-risk and binge drinking, particularly among individuals without a high school degree, White individuals, and older adults.[11] The second study showed the null effect of minimum wage on heavy drinking,[10] but higher minimum wages reduced binge drinking for women; while a third study[9] focusing on

Table 3. Coefficients for interactions between pre/post-indicator and treatment group status for six outcomes derived from the AUDIT instrument, adjusting for the person- and time-fixed effects, household size, health condition, marital status, home ownership status, and disability.

	Without spillover effects				With 10% spillover effects (ie, people up to 110% of the minimum wage are included in the treatment group)			
	Coef.	95% CI		*p*	Coef.	95% CI		*p*
Medium risk for alcohol use disorder[a]	0.011	−0.023	0.045	0.530	0.027	−0.005	0.059	0.100
High risk for alcohol use disorder[b]	0.011	−0.005	0.027	0.167	0.019	0.002	0.036	0.028
Addiction likely[c]	0.010	0.000	0.019	0.045	0.012	0.003	0.020	0.007
Hazardous consumption[d]	0.042	0.004	0.081	0.032	0.036	0.002	0.071	0.041
Alcohol dependence[e]	0.032	−0.003	0.066	0.074	0.032	0.000	0.064	0.049
Alcohol-related Harm[f]	0.027	−0.001	0.054	0.057	0.043	0.014	0.072	0.004

All time-invariant covariates are adjusted by the person-fixed effect. Time-variant covariates, including household size, health condition, marital status, home ownership status, and disability are adjusted in the models.
[a]Medium risk: Alcohol Use Disorders Identification Test (AUDIT) score ≥8.
[b]High risk: AUDIT score ≥16.
[c]Addiction likely: AUDIT score ≥20.
[d]Hazardous: hazardous levels of consumption.
[e]Dependence: the presence or emergence of alcohol dependence.
[f]Harm: the manifestation of alcohol-related harm.

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youth found higher minimum wage was associated with a higher propensity to binge drink for young men but lower propensity to binge drink for young women. In contrast, our research suggests that the 2018 minimum wage hike in Korea heightened the risks of alcohol use disorder and harmful alcohol use across the low-income population, particularly among males and those aged 50-64. The first US study[11] and our study, both observed significant policy effects in the older age group (ie, age 50-64 in this study and 40-64 in the prior study), who are already at a heightened risk for harmful alcohol use and alcohol-related deaths.[48-51] The strong policy impact among those among these at-risk groups is likely related to their elasticity of demand related to alcohol consumption as discretionary income increases. Those with a predisposition to high-risk behaviors or with a history of addiction might find that a rise in income facilitates more regular access to alcohol. Such individuals are likely to exhibit greater elasticity in their demand for alcohol, making them more responsive to changes in their financial situation. While casual drinkers might diversify their expenditures with increased income, someone at risk of harmful alcohol use may allocate a substantial portion of their additional funds specifically to alcohol.[52,53]

Our study has some limitations. (1) Our small sample size may bias our results towards the null. However, using the panel data with person-fixed effect models allows us to compare before and after within the same person, which eliminates the problems associated with compositional change in prior studies, mitigates omitted variable bias, and provides stronger evidence for causal inference. (2) Participants might underreport their alcohol intake, thus inducing a bias whose directionality remains indeterminate. But prior studies show that AUDIT has been validated, compared with other screening or clinical diagnoses.[54-56] (3) Our study focuses on current employees, ignoring how changes in minimum wage might affect employment levels. Concentrating on the employed may miss broader economic effects of minimum wage changes.[5] However, prior studies suggest that increases in minimum wage have minimal effect on employment rates.[38,39,57] The unemployment rate remained around 3.7% preceding and succeeding the minimum wage increase in 2017 and 2018.[40] (4) We have observed some evidence of differential attrition in the treatment and control groups (ie, 25% in the former and 18% in the latter). Since people in low-SES groups and higher symptomatology typically have higher risk of being lost to follow-up (and this was observed more frequently in our treatment, or low-SES, group), the differential attrition likely led to a bias towards the null in our results. (5) Our data set does not contain detailed information on the proportion of earnings originating from overtime work. It may be possible that a minimum wage worker who regularly performs overtime work (paid at 1.5x the minimum wage rate)[58] may appear as someone who earns above minimum wage. Moreover, monthly salaried workers (~60% of the sample) normally cannot get overtime pay in Korea.[59] Our spillover effects analyses likely captured these regular overtime workers and appropriately classed them into the treatment group. (6) There may be a need for multiple-test corrections, given that we included 6 clinically relevant outcomes based on AUDIT. However, we did not conduct multiple-test corrections because our analysis already incorporates conservative approaches inherent in DID methods using individual-level fixed effects. Applying multiple-testing corrections in the context of evaluating a population-level intervention may increase the risk of type 2 errors. (7) Korean minimum wage workers are typically older, are more likely married, and own homes compared to minimum wage workers in North America.[21,60,61] These demographic differences may limit the generalizability of our findings to similar East Asian contexts but may be less relevant to North American context.

Policy implications

While the salutary impacts of increases in the minimum wage on health outcomes have been established,[62] our study suggests that an augmented minimum wage is concomitant with heightened risks of alcohol use disorders and harmful alcohol use. To mitigate the potential escalation in alcohol use disorders, wage augmentations should be synergized with a comprehensive, multifaceted approach to harmful alcohol use prevention, such as awareness, education, pricing, advertising, and labelling.[63] Prior studies have also shown that alcohol tax is an effective strategy to reduce alcohol-related problems[64] and may be considered following minimum wage increases as a potential mitigation strategy.

Supplementary material

Supplementary material is available at *American Journal of Epidemiology* online.

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Conflict of interest

The authors declare no conflicts of interest.

Data availability

Data are available in a public, open-access repository. This data can be accessed through the KOWEPS website at: https://www.koweps.re.kr:442/eng/data/data/list.do.

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